

IT 3883 Final Exam

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Sprint 1:

Requirements:

- The program will ask the user to enter a sentence describing one or more quantities of coins.
- Each coin description will contain:
A numeric quantity.
A coin denomination written as a word.
- Multiple coin descriptions will be separated by the word and.
- The program will take pennies, nickels, dimes, and quarters.
- The program will give these values:
Penny: 1 cent
Nickel: 5 cents
Dime: 10 cents
Quarter: 25 cents
- The program will calculate the total value of all coins.
- The program will convert the total number of cents into dollars.
- The result will be displayed with two digits after the decimal point.
- The program does not need to process ending punctuation.

Design:

Pseudocode:

Create a dictionary containing the value of each coin denomination

Ask the user to enter a sentence describing the coins

Convert the sentence to lowercase

Split the sentence wherever the word "and" appears

Set total cents equal to zero

FOR each coin description:

 Remove extra spaces

 Split the description into quantity and denomination

 Convert the quantity into an integer

 Find the value of the denomination

 Multiply the quantity by the denomination value

 Add the result to total cents

END FOR

Divide total cents by 100 to convert cents to dollars

Display the dollar amount with two decimal places

Testing:

1 penny and 2 nickels, Error: penny not found, Failed

4 dimes and 7 quarters, Passed

1 quarter and 3 pennies, Error: quarter not found, Failed

21 pennies and 17 dimes and 52 quarters, Passed

95 dimes and 73 quarters and 22 nickels and 36 pennies, Passed

1 nickel and 17 quarters, Error: nickel not found, Failed

21 nickels and 15 pennies, Passed

1 dime and 1 nickel and 1 penny and 1 quarter, Error: dime not found, Failed

Testing showed that my dictionary only contained plural denomination names like pennies and quarters. When the quantity was just one, you would use penny or quarter instead of pennies/quarters. The program tried to find the singular word in the dictionary and would give back a `KeyError`. It also assumed that every input had exactly two words in each coin description. Incorrectly formatted input could cause an `IndexError` or `ValueError`.

Sprint 2:

Requirements:

- The program will ask the user to enter a sentence describing one or more quantities of coins.
- Multiple coin descriptions will be separated by the word and.
- The program will recognize both singular and plural forms:
penny and pennies
nickel and nickels
dime and dimes
quarter and quarters
- The program will ignore extra spaces around the input.
- Coin values will be stored and calculated as cents to avoid floating-point calculations.
- The program will calculate the combined value of every coin.
- The result will be displayed as a dollar amount with exactly two decimal places.
- If the sentence has an invalid quantity, denomination, or format, the program will display an error message instead of crashing.

Design:

Pseudocode:

Create a dictionary containing singular and plural coin names

Assign each coin name its value in cents

Ask the user to enter the coin description

Remove spaces from the beginning and end

Convert the sentence to lowercase

Split the sentence using "and"

Set total cents equal to zero

TRY:

FOR each coin description:

Remove extra spaces

Split the description into words

IF the description does not contain exactly two words:

Display an invalid format message

Stop processing

Convert the first word into an integer

IF the quantity is negative:

Display an invalid quantity message

Stop processing

Store the second word as the denomination

IF the denomination is not in the coin dictionary:

 Display an unknown denomination message

 Stop processing

 Multiply the quantity by the denomination value

 Add the result to total cents

END FOR

Convert total cents into dollars

Display the total with two decimal places

CATCH an invalid number error:

 Display an error message

Testing:

1 penny and 2 nickels, Passed

4 dimes and 7 quarters, Passed

1 quarter and 3 pennies, Passed

21 pennies and 17 dimes and 52 quarters, Passed

95 dimes and 73 quarters and 22 nickels and 36 pennies, Passed

1 nickel and 17 quarters, Passed

21 nickels and 15 pennies, Passed

1 dime and 1 nickel and 1 penny and 1 quarter, Passed

10 pennies, Passed

0 quarters and 5 dimes, Passed

2 QUARTERS and 3 PENNIES, Passed

5 dollars, Error message, Unknown denomination message, Passed

three pennies, Error message, Invalid quantity message, Passed

-2 nickels, Error message, Negative quantity message, Passed

Empty input, Error message, Empty input message, Passed

The single denomination bug from Sprint 1 was corrected by adding both singular and plural forms of every supported coin to the dictionary. All required test cases now pass. Input validation was also added. The corrected program handles unknown denominations, nonnumeric quantities, negative quantities, empty input, and incorrectly formatted coin descriptions without crashing. No additional bugs were found during Sprint 2 testing.